International Journal of Technology and Design Education
https://doi.org/10.1007/s10798-026-10083-w

Functional pretence: what makes humans designers - a foundational perspective for design and design education

Cassius Kiani[1] · **Richard Kimbell** · **Niall Seery**[1]

Received: 27 June 2025 / Accepted: 15 March 2026
© The Author(s) 2026

Abstract
Almost everything in our modern world is artificial, from cruffins and fish flops, to penicillin and computers. Why is it that humans make these *things*? And why don't we see other non-human creatures (like chimpanzees) making these *things*? In this theoretical paper, we synthesize across anthropology, cognitive science, and developmental psychology to suggest that humans engage in what we describe as *functional pretence*, an intentional process where one interacts with what one (or another) imagines as real. Functional pretence is more than pretend play, it is the capacity to treat imagined states, communicated through declarative language, as grounds for real action. It's this capacity which enables humans to interact with an artificial world which contains all the information and knowledge generated by other humans. It removes the requirement for direct experience, and enables humans to take a *faithful leap* toward the novel and the unknown. By exploring the arguments which lead to functional pretence, we explore what is fundamental to the design process that leads to all the *things* in our modern world. We then apply this foundational perspective to design education, illustrating through classroom scenarios how functional pretence already operates in design learning and where interventions could improve outcomes.

Keywords Functional pretence · Design education · Imagination · Artefacts · Knowledge

The hard problem of cruffins

In this paper, we offer a theoretical perspective on design education, grounded in a synthesis across anthropology, cognitive science, and developmental psychology. We aim to construct a foundational argument, one which identifies what is uniquely human about artefact creation, and traces its implications back to how we teach design.

Design education literature is rich in process models, pedagogical strategies, and assessments of design capability (Kimbell & Stables, 2007; Cross, 2011). However, it less fre-

✉ Cassius Kiani
cassius@cassiuskiani.com

[1] Technological University of the Shannon (TUS), Athlone Campus, N37 HD68 Bunnavally, Athlone, Co. Westmeath, Ireland

Published online: 28 March 2026

quently asks a prior question: *what makes humans capable of design at all?* In this paper, we address this question by examining what separates human artefact creation from the tool-use observed in non-human creatures. Through our synthesis, we arrive at *functional pretence*, an intentional process where one interacts with what one (or another) imagines as real. Functional pretence depends on two prerequisites: (1) the capacity to imagine alternatives beyond direct experience, and (2) declarative communication which can encode and decode new and prior knowledge.

We position functional pretence at the intersection between individual cognition and socio-cultural participation. The capacity for functional pretence is held by all neurotypical humans and is often developed by school age (age 4 and above). However, its expression is fundamentally social. It requires declarative communication, shared intentions, and access to an artificial world built through prior human knowledge.

Nearly everything in our modern world is artificial (Simon, 1996). Laptops, toilets, and even natural parks have been made by humans yet nature itself is *competent without comprehension* (Dennett, 2018). When birds build nests or chimpanzees create simple tools, these activities require no comprehension from the creatures themselves (Suddendorf et al., 2022). Humans share similar behaviours, however, we also create complex, resource-intensive artefacts like cruffins, forks, and washlets. When we take evolution into account, it is not obvious why these artefacts exist at all.

Many fields have what are known as *hard problems*: problems where it is difficult to explain why things are the way they are (Walker, 2024). We propose that to take a foundational perspective on design, and on how to improve design education, we must solve the *hard problem of cruffins*. We must be able to explain why trivial artefacts like cruffins can exist side-by-side with artefacts like computers and penicillin, and why humans create them while our closest non-human relatives do not. If we can understand the differences which make a difference, we gain both a foundational understanding of design and a principled basis from which to improve design education.

In covering a broad literature, we explore this hard problem by answering three questions:

1. Can all humans make cruffins? Can non-human creatures do the same?
2. What is foundational to artefact creation? How does this lead to cruffins, and computers?
3. What ideas can we then explore to further design and design education?

Below we define several key terms used throughout this paper. These terms form a logical chain: declarative communication enables functional pretence, functional pretence enables faithful leaps, and faithful leaps enable interaction with the artificial world to create artefacts. In practice, this chain is recursive, functional pretence makes faithful leaps possible, and faithful leaps are the acts through which functional pretences are sustained (Table 1).

What makes artefact creation possible?

Artefacts, memes, and what humans know

Like modern societies, non-modern societies also produce artefacts. In Papua New Guinea, the Tauade create koteka, decorative gourds which cover their private areas, and their neigh-

Table 1 Key terms used throughout this paper as a helpful reference for the reader

Term	Definition
Declarative Communication	Communication which allows information about an organism's internal states (e.g. thoughts, feelings, design intentions) to be encoded and decoded by another organism. One cannot take a faithful leap toward another's position unless that position can be communicated
Functional Pretence	An intentional process where one interacts with what one (or another) imagines as real. Functional pretence enables humans to learn, create, and collaborate with no direct experience, by taking faithful leaps toward another's position
Faithful Leap	An act where one moves from their current position to the position shared by another. In design education, students take faithful leaps when they engage with a design brief by imagining themselves in someone else's shoes
Artificial World	Following Popper (1978), the artificial world contains theories, ideas, and artefacts made by humans across time. Humans access this world through functional pretence and declarative communication
Artefact	Anything created by humans, whether material (e.g. books, houses, cruffins) or abstract (e.g. marriage, theories)

bours the Asaro create ornate masks made from mud (Otto & Verloop, 1996; Hallpike, 2008). It is not uncommon for non-modern societies to lack the ideas and behaviours which modern societies assume are universal, such as large numbers, land ownership, and resource storage (Hallpike, 2018). The Brazilian Pirahã live in an area where food is in abundance, which means it is unconventional for the Pirahã to ration food in their possession (Everett, 2009, 2012). Even when the Pirahã have scarce resources, they continue to behave in unconventional ways:

> "Pirahã: Dan, can you buy us a canoe? Our canoes are rotten. Everett: Why don't you make a canoe? Pirahã: Pirahãs don't make (Brazilian) canoes. We don't know how. Everett: I know you can make a bark canoe; I have seen you do that. Pirahã: Bark canoes don't carry weight. One man, some fish, no more. Only Brazilian canoes are good. Pirahã canoes are no good." (Everett, 2009, pp. 75–77).

Based on this exchange, Everett (2009) takes some Pirahã to learn how to make Brazilian canoes. Despite learning how to make them, days later they insist that *Pirahãs don't make canoes*. Popper (2002), Deutsch (2011), and Marletto (2021) suggest that knowledge is what transforms systems. In this instance, it is not the knowledge to make canoes which transforms the Pirahã, it is the existing knowledge in Pirahã culture (what we might call a *meme*) that prevents them from making canoes.

A meme is knowledge which is transmitted through culture and replicates through imitation (Dawkins, 1976). The term was coined as a cultural analogue to the gene. Where

genes transmit biological information through reproduction, memes transmit knowledge through observation, instruction, and social learning (Dawkins, 1976; Blackmore, 1999). Like genes, memes face selection pressures. Genes are why all humans have opposable thumbs, and memes are why some societies choose chopsticks over forks. What makes memes especially relevant to our argument is that cognition is a finite resource, and ideas which are easy to grasp (whether helpful or harmful) require fewer resources to comprehend and fewer to propagate (Weng et al., 2012; Franconeri et al., 2013; Dennett, 2018). Memes do not merely shape behaviour, they shape whether an individual will engage in functional pretence at all. A meme that says "we don't make canoes" does not prevent the Pirahã from possessing the capacity to make canoes. However, it does prevent them from taking the faithful leap toward this knowledge.

The consequences compound when suppressive memes propagate. Memes, by definition, replicate, and a meme like "I can't design" does not remain with a single student. It spreads through peer groups, settles into classroom cultures, and can embed itself within curricula and institutions. Bandura (2023) describes how efficacy beliefs operate at both the individual and at the collective level. When educators share a belief that certain students cannot design, this belief becomes institutional. Based on what we have established, memes like "I can't design" or "design is for the gifted" are fallacies. If all humans possess the capacity for functional pretence, and functional pretence is foundational to artefact creation, then "I can't design" is not a statement about capacity, it is a meme which suppresses it. In this sense, memes are active forces which either open, or close, the spaces in which faithful leaps can occur. Identifying suppressive memes as fallacies is itself a corrective meme, one which, if it propagates, can begin to reopen these spaces.

The differences which make a difference

Chimpanzees are our closest living relatives, and we continue to learn that there are more overlaps than there are differences (Prüfer et al., 2012; Kuhlwilm et al., 2016). Goodall (2000) documents how many once "unique" human behaviours are also seen in chimpanzees. Tomasello (2022) describes how chimpanzees can use complex tools and engage in collaboration to make these tools. Whiten et al. (2021) highlights how chimpanzees will develop new tools as their environments change. By focusing on chimpanzees, any foundational ideas we develop which cannot be seen in our closest relatives are more likely to be *differences which make a difference*.

What is really required to create cruffins? Cruffins require resources from across the globe, knowledge from various cultures, coordination to divide the labour, and a space where imagination becomes material. Cruffins are more than physical things, their existence represents the knowledge, ideas, and explanations which permit their existence (Deutsch, 2011; Marletto, 2021). Flour represents knowledge that certain plants can be milled into consumable resources. Wheat does not start as flour, and for flour to exist, we require knowledge which is not available to humans at birth (Dehaene, 2020). Knowledge to create flour is not encoded in human DNA, in much the same way that the knowledge to gather honey does not exist in chimpanzees at birth.

Knowledge requires functional pretence What makes knowledge generation unique to humans is that we are able to engage in what we call *functional pretence*, an intentional process where one interacts with what one (or another) imagines as real. It is not possible to know how to mill flour, or how to bake cruffins, through observation alone. However, it is possible to operate under a functional pretence to interact with the possible ideas and explanations (communicated by another) which make this possible.

Even the Pirahã, who refuse to believe anything which has not been directly witnessed, engage in functional pretence (Everett, 2009). Ostracising an individual requires more than one person to harbour a disdain, it requires groups to communicate, and for others who did not witness the event to operate under the same ideas. Individuals who engage in this process must be able to imagine that what has taken place is real, and then act on this new knowledge.

Functional pretence is not the same as pretence. Pretence activities take place in the present and require direct experience. Rough-and-tumble play in chimpanzees and meerkat hunting are simple examples (Panksepp, 1993; Thornton & McAuliffe, 2006; Panksepp & Lahvis, 2007). In a design classroom, a student who has never met an elderly person can still design a mobility aid for them through functional pretence.

Functional pretence requires declarative communication Communication is by no means unique to humans, nor are symbols or symbolism (Rumbaugh et al., 2009; Pepperberg, 2021). Elephants have been shown to call other elephants by name, and chimpanzees howl to alert others to danger (Poole et al., 1988; Dezecache et al., 2019). However, *declarative communication*, communication which allows information to be encoded about an organism's internal states (e.g. thoughts, feelings, design intentions) and then decoded by another organism, does appear to be unique to humans.

Declarative communication can be verbal, symbolic, or anything in-between, as long as the information can be encoded and decoded reliably. While chimpanzees can learn sign language, research suggests only better-than-chance odds that chimpanzees engage in declarative communication (Ristau & Robbins, 1982; Gardner et al., 1991; Terrace, 2019; Hopkins et al., 2022). Instead, chimpanzees engage in imperative communication, communications which commands or instructs an action (e.g. *feed me*). Declarative communication is necessary for functional pretence, because it is not possible to take a faithful leap toward someone else's position unless they can communicate that position.

When a design educator presents a brief (e.g. design a product that helps elderly people feel less isolated) students must take a faithful leap from their own experience into the imagined experience held by another, and take designerly actions as if that situation were real. This is functional pretence in action, and it's made possible by the declarative communication between student and educator.

How functional pretence builds the artificial world Declarative communication helps individuals organise into larger groups where what one creates or imagines can be taken further through organised structures tribes, companies, research laboratories. Tomasello (2022) describes this as shared intentionality, where groups share the same attention, focus, and outcomes. When humans share intentions under the same functional pretence, artefacts

accumulate and the artificial world grows (Tennie et al., 2009). Following Popper (1978), we use the artificial *world* to describe the accumulated theories, ideas, and artefacts made by humans across time. Non-human creatures share the natural and experiential worlds with us, however only humans can take ideas from the experiential world and share them through declarative communication. Humans born today can take faithful leaps through functional pretence to learn, explore, and contribute to this artificial world. Whereas we've yet to see any non-human creature contribute to this artificial world, or to begin building their own (Seed & Byrne, 2010).

Humans can also use functional pretence to generate knowledge through what Searle (2001) describes as different *fits*. Most artefact creation is a mind-to-world fit, the idea starts in the mind before it ends up in the world. Physics follows a world-to-mind fit, we shape our ideas to match how nature works. Functional pretence also enables mind-to-mind fits, where one takes a faithful leap from one's own position to the position held by another.

In a design classroom, all three fits operate simultaneously. When students research a user's needs (world-to-mind), imagine a solution (mind-to-world), and align their design intentions with a teammate's vision during a group project (mind-to-mind), they exercise the full range of functional pretence. In this frame, design studios can be seen as structured environments for practising faithful leaps across all three fits.

Functional pretence in development and in the classroom

Developmental foundations

Given that children are unable to build companies or join research labs, it is worth posing the question, when does functional pretence begin to appear in humans?

Protoconversations, non-verbal exchanges where children respond to sounds, faces, and intentions, appear from as young as six weeks (Tomasello, 2008). During protoconversations, children take faithful leaps during each interaction to attempt to align themselves with their conversation partner (Bateson, 1975; Yoo et al., 2018). These interactions support emotional and cognitive development, teaching infants the relationship between their actions and the responses of others (Greenspan & Shanker, 2004). Around 18 months, children engage in pretend play, transforming simple objects like sticks and stones into other artefacts like swords (Nielsen & Dissanayake, 2004). Children also begin to engage with symbolism around a similar time (Adamson et al., 2004; DeLoache, 2004). By age 3, pretend play expands from individual games to larger collaborative games which include transforming toys into characters and involving other children in behaviours which resemble functional pretence (Parker et al., 2015).

Based on research from Tomasello (2022), it is at age 4 where the starkest cognitive contrasts appear between humans and chimpanzees. Counterfactual imagination, the ability to generate imagined alternatives based on experience (e.g. *if I did this… then…*) is a notable example (Byrne, 2016; Hoeck et al., 2013; McCormick et al., 2018). Once counterfactual imagination develops, school-age children possess all the prerequisites to engage in functional pretence.

Functional pretence in the design classroom: three scenarios

To make the relationship between functional pretence and design education concrete, we present three classroom scenarios, including what occurs when functional pretence breaks down.

Scenario 1: The community garden brief A class of 12-year-olds is given a design brief, a local community centre needs seating for its garden that elderly visitors can use comfortably. To engage meaningfully, students must imagine themselves into the position of an elderly person they have likely never met. Someone with reduced mobility, different comfort requirements, different aesthetic preferences, and treat this imagined person's needs as real constraints on their design decisions. They must then communicate their design intentions through sketches, models, and verbal explanations, encoding their internal reasoning into forms that others can decode and critique. Kimbell and Stables (2007) describe this as the interaction between imaging and modelling inside the head, and confronting the reality outside the head.

Scenario 2: Novel engineering (primary) McCormick and Hynes (2012; see also Milto et al., 2020) document a programme called Novel Engineering, where primary school children read literature and then identify engineering problems faced by fictional characters. In one instance, students read *The Swiss Family Robinson* and designed a shelter to keep the characters cool under the hot sun. In another, students designed a hearing aid for the main character in *El Deafo*, and styled it as a fashion accessory. In each case, the functional pretence chain is visible: students decoded the narrative (declarative communication), imagined the character's problem as their own (faithful leap), treated the fictional constraints as real (functional pretence), and created a material artefact in response. When students take a fictional scenario seriously enough to generate workable solutions, they exercise the same cognitive architecture that underlies all human artefact creation.

Scenario 3: The studio crit (higher education) Schön's (1983) documentation of the Quist–Petra studio exchange provides a well-known illustration. Petra, an architecture student, is stuck on some early sketches. Quist, her studio lead, places tracing paper over her drawings and begins to redraw, narrating his design thinking aloud. In our terms, Quist makes visible the imagined future he is working toward, creating the conditions for Petra to take a faithful leap from her position to his. The educational power lies in the fact that his declarative communication enables Petra to enter and work within this functional pretence. However, this exchange depends on Petra's willingness to leap. If a suppressive meme prevents this, the same declarative communication meets a closed door. While the studio critique is perhaps the most iconic format in design education, it is only as effective as the student's capacity to engage in the functional pretence it requires.

When functional pretence breaks down

Functional pretence becomes clearest when it fails. Consider a student given the community garden brief from Scenario 1 who designs a bench that is low to the ground, brightly coloured, and made from hard plastic (essentially, a bench the student themselves would

enjoy). When asked about the elderly user's needs, they respond: "I think this would be comfortable." In this framing, the student has not taken the faithful leap, and they are designing under their own pretence, rather than under the pretence set by the brief.

If the limitation is a failure of functional pretence, then the intervention is not to teach more about materials or ergonomics. The intervention is to strengthen the student's capacity for faithful leaps. Methods such as empathy maps, or "a day in the life" exercises then become interventions to strengthen functional pretence rather than design processes.

Summary

In this paper, we have explored a broad literature to answer three questions.

1. Can all humans make cruffins? Can non-human creatures do the same? We address this question by suggesting that all humans could make cruffins. Anything which exists within the universe can be created or replicated by any human, as long as we are able to create (or find) the knowledge to do so. While there may be memes (cultural ideas) or gaps in knowledge which prevent this, in principle all humans could make cruffins. As long as the knowledge exists, humans can engage in functional pretence to take faithful leaps toward it.

As for non-human creatures, we have highlighted that declarative communication is required to engage in functional pretence and to interact with the artificial world where the knowledge to create cruffins exists. There is limited evidence to suggest this is possible, or probable, in non-human creatures.

2. What is foundational to artefact creation? How does this lead to cruffins, and computers? We suggest that functional pretence is foundational for artefact creation. Functional pretence allows humans to interact with what one (or another) imagines as real, which in turn allows interaction with the artificial world containing human ideas and knowledge. Provided knowledge exists in books, stories, videos, or similar, there is no need to rediscover or recreate it. One can simply take a faithful leap from one's current position to the position shared by another. As artefacts become more complex, they require more specific knowledge and more faithful leaps. For both cruffins and computers, we require an artificial world where specific knowledge exists, alongside functional pretence to interact with and take faithful leaps toward this knowledge.

3) What ideas can we then explore to further design and design education? Functional pretence contains several ideas we can explore to improve design and design education. We outline three:

Improving declarative communication improves design. Interacting with the artificial world and the knowledge within it requires declarative communication. This means that design educators should attend as carefully to *how* students communicate their design reasoning as to the artefacts they produce. Students who can articulate *why* they made a design decision, not merely *what* they decided, enable richer critique and faster

iteration. Interventions which improve introspection, emotional intelligence, and the capacity to encode design reasoning for others may therefore yield measurable improvements in design outcomes. Schön's (1983) Quist–Petra exchange highlights that educational power lies in the declarative externalisation of an imagined alternative, making it visible and shareable.

Design education can start early. Cognitive prerequisites for functional pretence are in place by school age. There is no reason to delay engaging children in design or designerly activities from the moment they enter education. Earlier interventions which develop these behaviours are likely to be beneficial. Some interventions may even include miniaturising the structures which benefit most from functional pretence for example, classroom design studios that simulate client-designer relationships, or student-led "companies" that must pitch and iterate. Making these structures suitable, engaging, and useful for school-age children (age 4 and above) is the challenge.

Useful memes matter more than talent. One final (and familiar) idea for design educators is that we must instill in our students the idea that they are able to design and that they can design. Functional pretence gives this familiar intuition theoretical grounding: the Pirahã example demonstrates that culturally transmitted beliefs about what one *does* and *doesn't do* can override actual capability. Whether we arrive at this through anthropological literature or through Bandura's (2023) Social Cognitive Theory, the implication is the same: we must help empower our students with useful memes rather than useless ones. In practical terms, this means attending to the identity narratives present in our classrooms. Do students see themselves as people who design, or as people who follow instructions?

Suggestions for further research

We encourage researchers and design educators to explore foundational perspectives on design. Foundational perspectives can help guide new ideas, interventions, and validation methods which underpin design education. As an example, functional pretence provides a foundational perspective to assess why a promising educational intervention could succeed, or fail to meet expectations.

Given our hypothesis on declarative communication, research into methods which improve declarative communication in both educators and students may yield notable results. While there is a broad literature which discusses emotional intelligence, introspection, and communication in education, functional pretence brings a novel perspective on how design educators can engage with this literature with design outcomes in mind.

In another paper, we discuss the idea that how humans choose to dedicate their limited resources (e.g. what they choose to design) stems from a phenomenon we have described as *tension* (Kiani et al., 2025). Research which maps functional pretence and tension to both the design process and classroom interventions will likely lead to productive results. Assuming that both functional pretence and tension are foundational toward the design process, we would expect interventions which bridge these two ideas to improve learning outcomes alongside the artefacts students create.

Perhaps the most important research which will come from this paper is additional work which explores how these foundational perspectives map to successful interventions across

other fields. Our hope is that functional pretence, alongside our other foundational perspectives on design, will help researchers and educators to find (and bring back) successful interventions from other fields into design education. We believe this is possible because these foundational perspectives bring more refined search spaces, and new lenses, to make it easier to identify *why* something is useful.

Given we now live in a world which, while still resource-constrained, contains more knowledge and technologies than ever before, it seems valuable to explore foundational perspectives on design. We believe foundational perspectives will help bring forward our students, our field, and our species.

Funding Open Access funding provided by the IReL Consortium. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Adamson, L. B., Bakeman, R., & Deckner, D. F. (2004). The development of symbol-infused joint engagement. *Child Development*, *75*(4), 1171–1187.

Bandura, A. (2023). *Social Cognitive Theory: An Agentic Perspective on Human Nature*. Edited by D. Cervone. Hoboken, NJ: John Wiley & Sons.

Bateson, M. C. (1975). Mother-infant exchanges: the epigenesis of conversational interaction. *Annals of the New York Academy of Sciences*, *263*(1), 101–113.

Blackmore, S. (1999). *The Meme Machine*. Oxford University Press.

Byrne, R. (2016). Counterfactual Thought. *Annual Review of Psychology*, *67*, 135–157.

Cross, N. (2011). *Design Thinking: Understanding How Designers Think and Work*. Berg.

Dawkins, R. (1976). *The Selfish Gene*. Oxford University Press.

Dehaene, S. (2020). *How We Learn: The New Science of Education and the Brain*. Penguin UK.

DeLoache, J. S. (2004). Becoming symbol-minded. *Trends in Cognitive Sciences*, *8*(2), 66–70.

Dennett, D. C. (2018). *From Bacteria to Bach and Back: The Evolution of Minds*. W.W. Norton & Company.

Deutsch, D. (2011). *The Beginning of Infinity: Explanations That Transform the World*. Allen Lane.

Dezecache, G., Crockford, C., & Züberbühler, K. (2019). The development of communication in alarm contexts in wild chimpanzees. *Behavioral Ecology and Sociobiology*, *73*, 104.

Everett, D. L. (2009). *Don't Sleep, There Are Snakes: Life and Language in the Amazonian Jungle*. Knopf Doubleday Publishing Group.

Everett, D. L. (2012). What does Pirahã grammar have to teach us about human language and the mind? *Wiley Interdisciplinary Reviews: Cognitive Science*, *3*(6), 555–563.

Franconeri, S. L., Alvarez, G. A., & Cavanagh, P. (2013). Flexible cognitive resources: competitive content maps for attention and memory. *Trends in Cognitive Sciences*, *17*(3), 134–141.

Gardner, R. A., Gardner, B. T., & Van Cantfort, T. E. (1991). Teaching sign language to chimpanzees. *International Journal of Primatology*, *12*, 303–307.

Goodall, J. (2000). *the Shadow of Man*. Houghton Mifflin Harcourt.

Greenspan, S. I., & Shanker, S. G. (2004). *The First Idea: How Symbols, Language, and Intelligence Evolved from Our Primate Ancestors to Modern Humans*. Da Capo.

Hallpike, C. R. (2008). *How We Got Here: From Bows and Arrows to the Space Age*. AuthorHouse.

Hallpike, C. R. (2018). *Ship of Fools: An Anthology of Learned Nonsense about Primitive Society*. Castalia House.

Hoeck, N. V., Ma, N., Ampe, L., Baetens, K., Vandekerckhove, M., & Van Overwalle, F. (2013). Counterfactual thinking: an fMRI study on changing the past for a better future. *Social Cognitive and Affective Neuroscience*, *8*(5), 556–564.

Hopkins, W. D., Mulholland, M. M., Mareno, M. C., Webb, S. J., & Schapiro, S. J. (2022). Neuroanatomical correlates of individual differences in the object choice task in chimpanzees. *Frontiers in Psychology*, *13*, 1057722.

Kiani, C., Kimbell, R., & Seery, N. (2025). A Tension is All You Need: A Theory on Why Humans Make Things. *International Journal of Technology and Design Education*. https://doi.org/10.31235/osf.io/dba2p_v1

Kimbell, R., & Stables, K. (2007). *Researching Design Learning: Issues and Findings from Two Decades of Research and Development*. Springer.

Kuhlwilm, M., de Manuel, M., Nater, A., Greminger, M. P., Krützen, M., & Marques-Bonet, T. (2016). Evolution and demography of the great apes. *Current Opinion in Genetics & Development*, *41*, 124–129.

Marletto, C. (2021). *The Science of Can and Can't: A Physicist's Journey Through the Land of Counterfactuals*. Penguin Books Limited.

McCormick, C., Rosenthal, C. R., Miller, T. D., & Maguire, E. A. (2018). Mind-Wandering in People with Hippocampal Damage. *The Journal of Neuroscience*, *38*(11), 2745–2754.

McCormick, M., & Hynes, M. (2012). *Engineering in a Fictional World: Early Findings from Integrating Engineering and Literacy*. ASEE Annual Conference & Exposition, San Antonio, Texas. https://doi.org/10.18260/1-2--21307

Milto, E., Portsmore, M., McCormick, M., Watkins, J., & Hynes, M. (2020). *Novel Engineering, K–8: An Integrated Approach to Engineering and Literacy*. NSTA.

Nielsen, M., & Dissanayake, C. (2004). Pretend play, mirror self-recognition and imitation: A longitudinal investigation. *Infant Behavior and Development*, *27*(3), 342–365.

Otto, T., & Verloop, R. J. (1996). The Asaro Mudmen: Local Property, Public Culture? *The Contemporary Pacific*, *8*(2), 349–386.

Panksepp, J. (1993). Rough and tumble play: A fundamental brain process. In K. MacDonald (Ed.), *Parent–Child Play* (pp. 147–184). State University of New York.

Panksepp, J. B., & Lahvis, G. P. (2007). Social reward among juvenile mice. *Genes Brain and Behavior*, *6*(7), 661–671.

Parker, J. G., Rubin, K. H., Erath, S. A., Wojslawowicz, J. C., & Buskirk, A. A. (2015). Peer Relationships, Child Development, and Adjustment. In D. Cicchetti, & D. J. Cohen (Eds.), *Developmental Psychopathology* (pp. 419–493). Wiley.

Pepperberg, I. M. (2021). Nonhuman and Nonhuman-Human Communication. *Frontiers in Psychology*, *12*, p647841.

Poole, J. H., Payne, K., LangbauerJr, W. R., & Moss, C. J. (1988). The social contexts of some very low frequency calls of African elephants. *Behavioral Ecology and Sociobiology*, *22*(6), 385–392.

Popper, K. R. (1978). *Three Worlds. The Tanner Lecture on Human Values*. Retrieved from https://tannerlectures.utah.edu/_resources/documents/a-to-z/p/popper80.pdf

Popper, K. R. (2002). *Conjectures and Refutations: The Growth of Scientific Knowledge*. Routledge.

Prüfer, K., Munch, K., Hellmann, I., et al. (2012). The bonobo genome compared with the chimpanzee and human genomes. *Nature*, *486*(7404), 527–531.

RistauC.A., & Robbins, D. (1982). Language in the Great Apes: A Critical Review. *Advances in the Study of Behavior*, *12*, 141–255.

Rumbaugh, D. M., Savage-Rumbaugh, E. S., & Taglialatela, J. P. (2009). Language: Nonhuman Animals. *Encyclopedia of Neuroscience* (pp. 367–371). Academic.

Schön, D. A. (1983). *The Reflective Practitioner: How Professionals Think in Action*. Basic Books.

Searle, J. R. (2001). *Rationality in Action*. MIT Press.

Seed, A., & Byrne, R. (2010). Animal Tool-Use. *Current Biology*, *20*(23), R1032–R1039.

Simon, H. A. (1996). *The Sciences of the Artificial* (3rd ed.). MIT Press.

Suddendorf, T., Bulley, A., & Redshaw, J. (2022). *The Invention of Tomorrow: A Natural History of Foresight*. Basic Books.

Tennie, C., Call, J., & Tomasello, M. (2009). Ratcheting up the ratchet: on the evolution of cumulative culture. *Philosophical Transactions of the Royal Society B*, *364*(1528), 2405–2415.

Terrace, H. S. (2019). *Why Chimpanzees Can't Learn Language and Only Humans Can*. Columbia University.

Thornton, A., & McAuliffe, K. (2006). Teaching in wild meerkats. *Science*, *313*(5784), 227–229.

Tomasello, M. (2008). *Origins of Human Communication*. MIT Press.

Tomasello, M. (2022). *The Evolution of Agency: Behavioral Organization from Lizards to Humans*. MIT Press.

Walker, S. I. (2024). *Life As No One Knows It: The Physics of Life's Emergence*. Hachette UK.

Weng, L., Flammini, A., Vespignani, A., & Menczer, F. (2012). Competition among memes in a world with limited attention. *Scientific Reports*, *2*, 335.

Whiten, A., Harrison, R. A., McGuigan, N., Vale, G. L., & Watson, S. K. (2021). Collective knowledge and the dynamics of culture in chimpanzees. *Philosophical Transactions of the Royal Society B*, *377*(1843), p20200321.

Yoo, H., Bowman, D. A., & Oller, D. K. (2018). The Origin of Protoconversation. *Frontiers in Psychology*, *9*, 1510.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.